

Home Practice Kit — Assembly Guide (for the coach)

Assemble one kit per student. First time takes about an hour; after that it's just refills. Hand out at Session 1, along with the [Home Lab Safety Agreement](#).

Shopping list (makes ~4 kits)

Powders — all from a grocery store:

- Baking soda
- Corn starch
- Table salt (iodized is fine)
- Granulated sugar
- Powdered (confectioner's) sugar
- Epsom salt (pharmacy — plain, unscented)
- Cream of tartar (spice aisle)
- Citric acid (canning aisle, or "sour salt")
- Calcium carbonate — sidewalk chalk crushed to powder, or a plain calcium supplement crushed, or "garden lime"
- Play sand (a small bag; rinse and dry it first)
- *(optional extras for later: baking powder, unflavored gelatin, MSG, sodium-acetate hand-warmer beads, washing soda / "super washing soda")*

Reagents:

- Tincture of iodine (pharmacy antiseptic — the 2% "tincture," not povidone/ Betadine, and not "decolorized" iodine). One small bottle per kit.
- Red cabbage (one head makes indicator for the whole team, several times).

Containers:

- Small zip bags (snack size) or mini condiment cups with lids — ~11 per kit
- One 1-oz dropper bottle per kit for the iodine
- One 2-oz bottle per kit for the cabbage indicator
- A sandwich-size bag or small box to hold each kit
- Sticky labels

Portioning the powders

Into each labeled bag/cup, put about **2–3 tablespoons**. That's plenty for many practice runs. Label each with the **name** (the kids already know them — this isn't a blind set; blind "unknown" sets are made fresh at practice).

Label set: baking soda · corn starch · table salt · granulated sugar · powdered sugar · Epsom salt · cream of tartar · citric acid · calcium carbonate · sand plus a couple of **blank numbered bags** ("Unknown 1", "Unknown 2") the parent can fill for a blind quiz.

Making red-cabbage indicator

1. Chop about $\frac{1}{4}$ of a red cabbage.
2. Cover with water in a pot, simmer 10–15 minutes until the water is deep purple. (Or pour boiling water over chopped cabbage in a bowl and wait 30 min.)
3. Strain out the cabbage. The purple liquid is your indicator.
4. Cool, then fill each kit's 2-oz dropper bottle. Refrigerate the rest; it keeps about a week (freeze in an ice tray for longer — thaw a cube as needed).

Color guide to print small and tuck in each kit:

Color	Approx. pH
red	2
pink	4
purple	7 (neutral)
blue-purple	8
blue-green	9–10
green	11
yellow	12+

What the family supplies (put this list in the kit)

- White vinegar
- Water
- A magnet (fridge magnet is fine)
- Clear cups, or a muffin tin / ice-cube tray
- Toothpicks, a spoon
- Paper towels
- Safety goggles or well-fitting glasses (if the family doesn't have goggles, add a cheap pair to the kit)

What is deliberately NOT in the kit

No strong acid or base. No hydrogen peroxide (fine for the family to have, but not part of the kit — kids shouldn't be doing the blood-catalase demo unsupervised). No heat source, no flame, nothing for burn tests. Those all stay at in-person practice.

Refills

Kids will run low on iodine and cabbage indicator first. Bring a refill bottle of each to every meeting. Powders last most of the season; top up the ones they use most (baking soda, corn starch, citric acid).

Cost

Roughly \$8–12 per kit, most of it one-time (the powders and cabbage serve the whole season). The dropper bottles are the main recurring cost if kids lose them.